



Aais the Adventurer's Quest

Problem ID: c

Time Limit: 1 second
Memory Limit: 256 MB

Aais the Adventurer is traveling on the 2D coordinate plane. He is currently at point $(0, 0)$ and wants to reach point (x, y) . In one move, he chooses an integer d such that $0 \leq d \leq k$ and jumps d spots forward in the direction he is facing.

Initially, Aais is facing the positive x-direction. After every move, he alternates between facing the positive x-direction and the positive y-direction (i.e., he will face the positive y-direction on his second move, the positive x-direction on his third move, and so on).

What is the minimum number of moves he must perform to land on point (x, y) ?

Input

The first line contains an integer t ($1 \leq t \leq 10^4$) — the number of test cases.

Each test case contains three integers x, y, k ($0 \leq x, y \leq 10^9, 1 \leq k \leq 10^9$) — the destination coordinates and the maximum jump distance.

Output

For each test case, output the number of jumps Aais needs to make on a new line.

Sample Input 1

```
3
9 11 3
0 10 8
1000000 100000 10
```

Sample Output 1

```
8
4
1999999
```

Sample Input 2

```
5
15 12 5
7 9 4
1 1 1
20 30 6
100 100 10
```

Sample Output 2

```
6
6
2
10
20
```